

Exercise

Take a closer look at the following graphics and address these questions:

- Does the visualization serve to **explore or to explain**?
- Is the visualization **easy to understand**?
- What is the **main message** you learn from the graphic? (if any)
- Are differences, patterns, or trends **easily recognizable**?
- How do you feel about the **overall aesthetics and elegance** of the design?

Collect three things you notice, no matter if positive or negative.

- How could you fix the details you dislike?



I feel that I can...

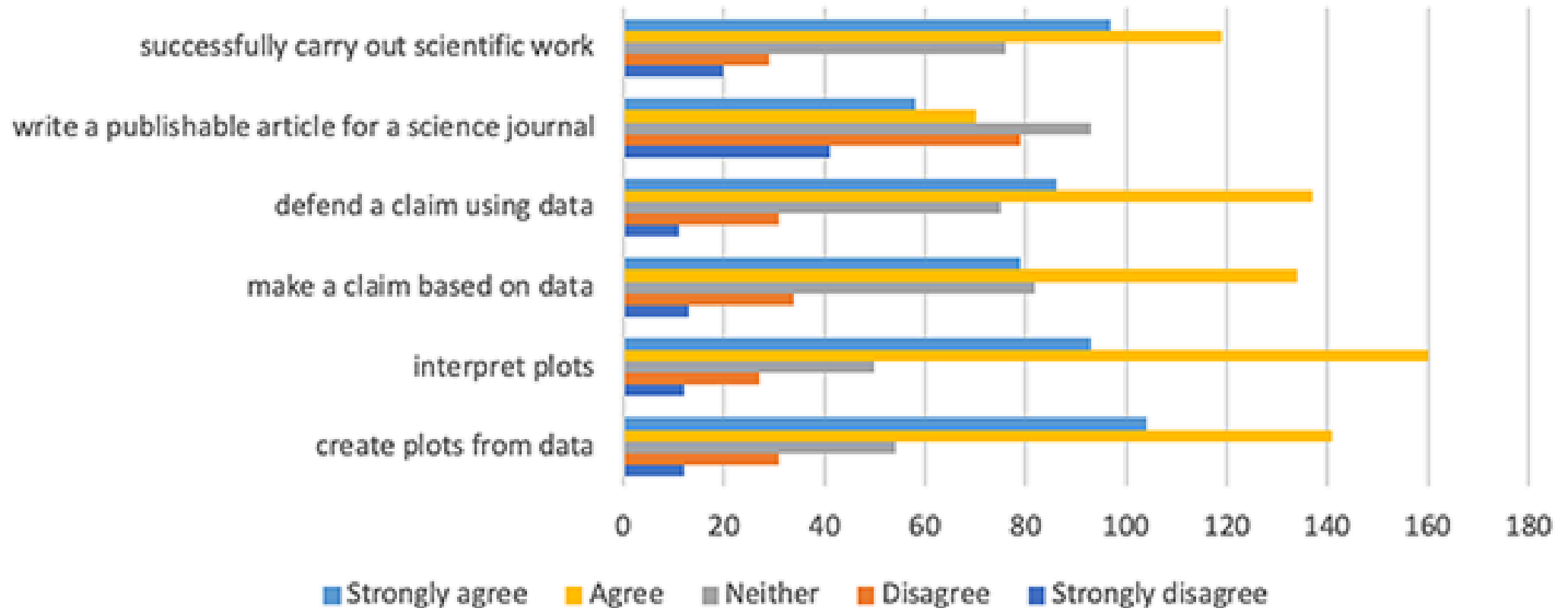


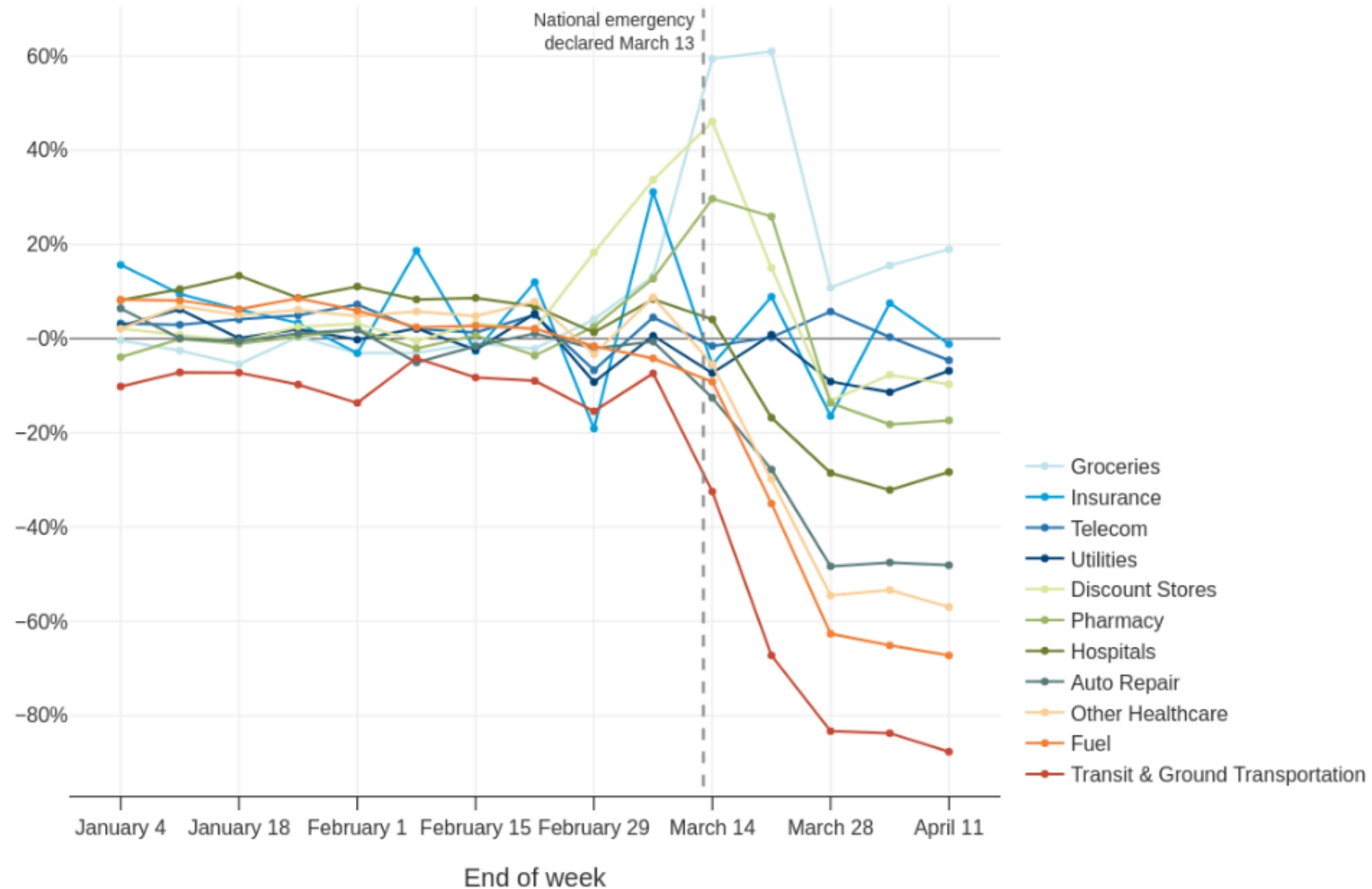
Figure 6

Histogram of responses to questions regarding perceived data analysis and reporting skills (N = 490).

Crowston, Mitchell & Østerlund (2019) *Citizen Science*



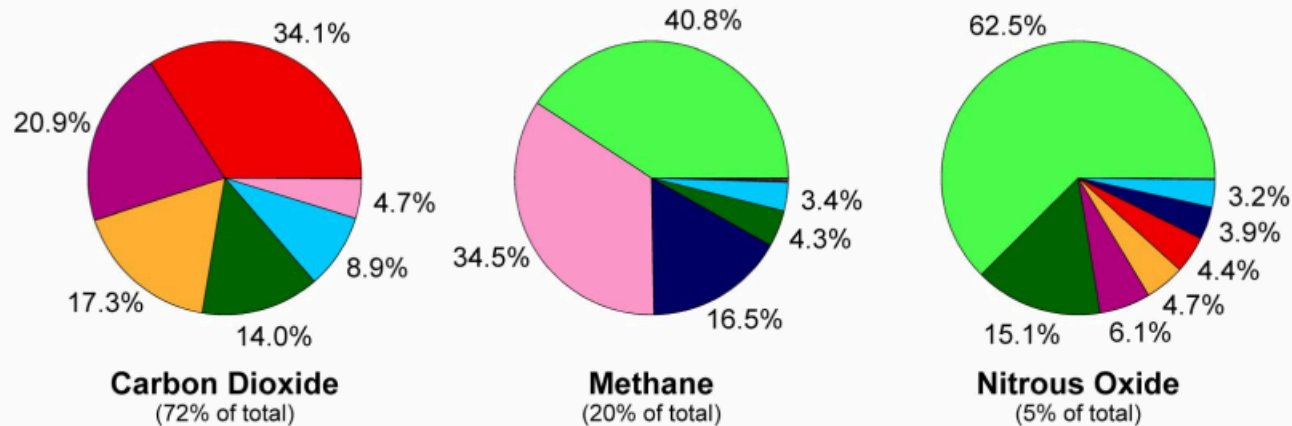
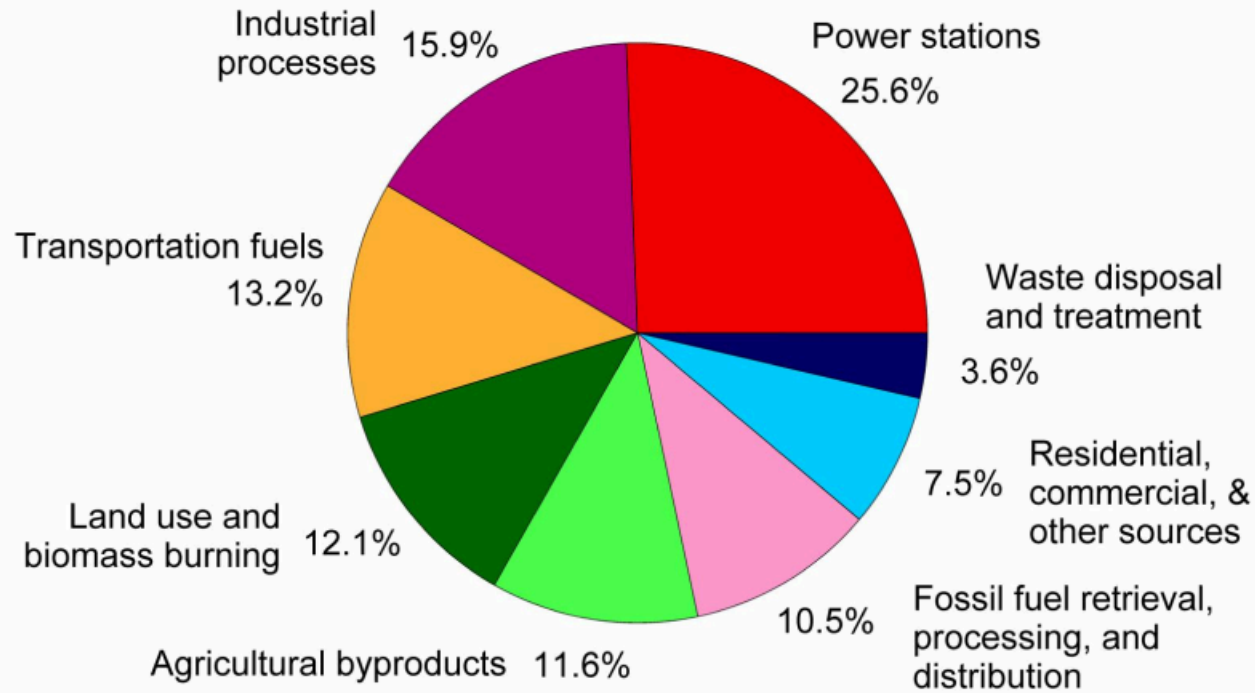
Year-over-year percent change in spending by essential category



Source: JPMorgan Chase Institute



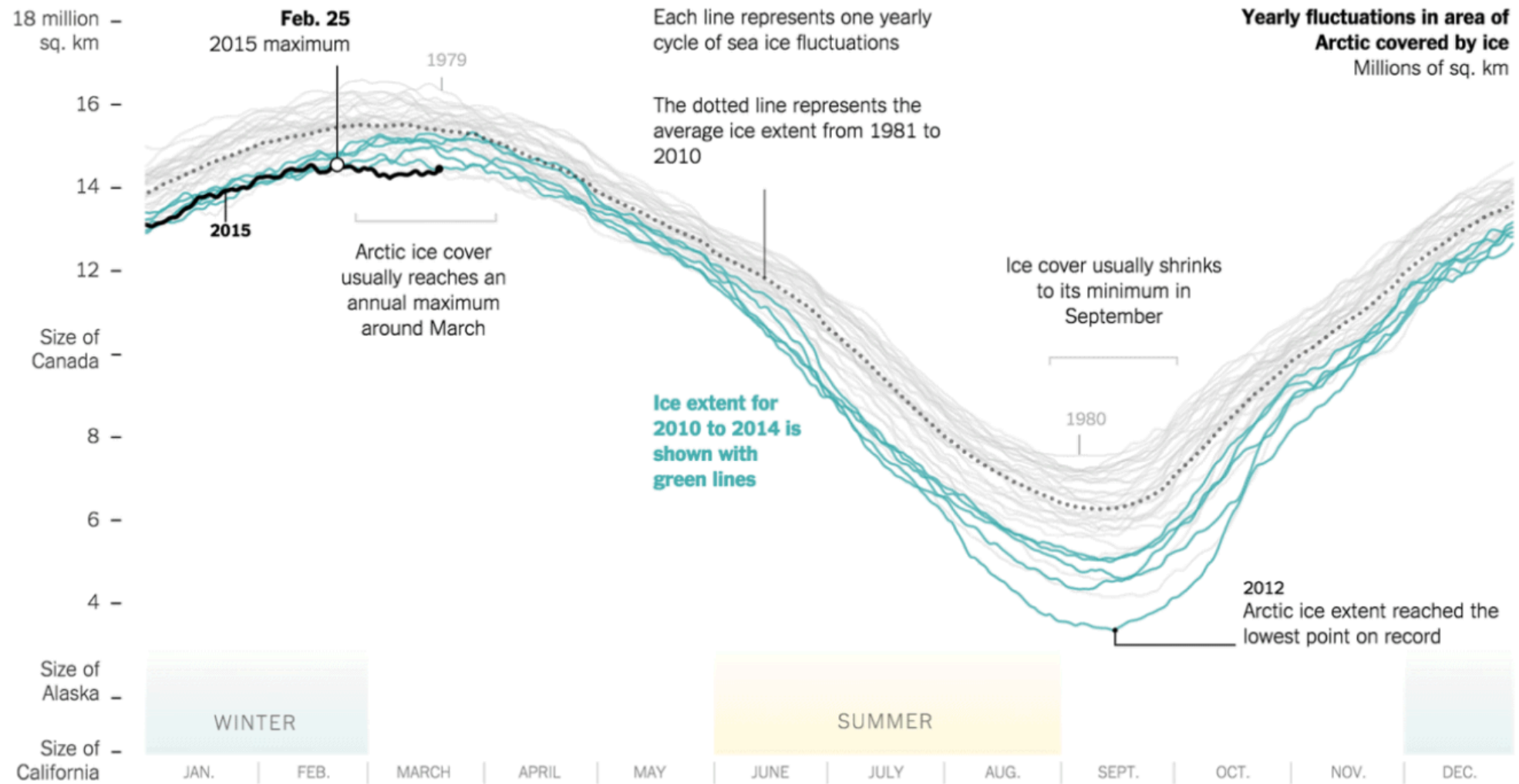
Annual Greenhouse Gas Emissions by Sector



Emissions data for the year 2010 derived from the Emissions Database for Global Atmospheric Research (EDGAR) v4.2 FT2010

Source: Dr. Robert Rohde ([Tweet](#))



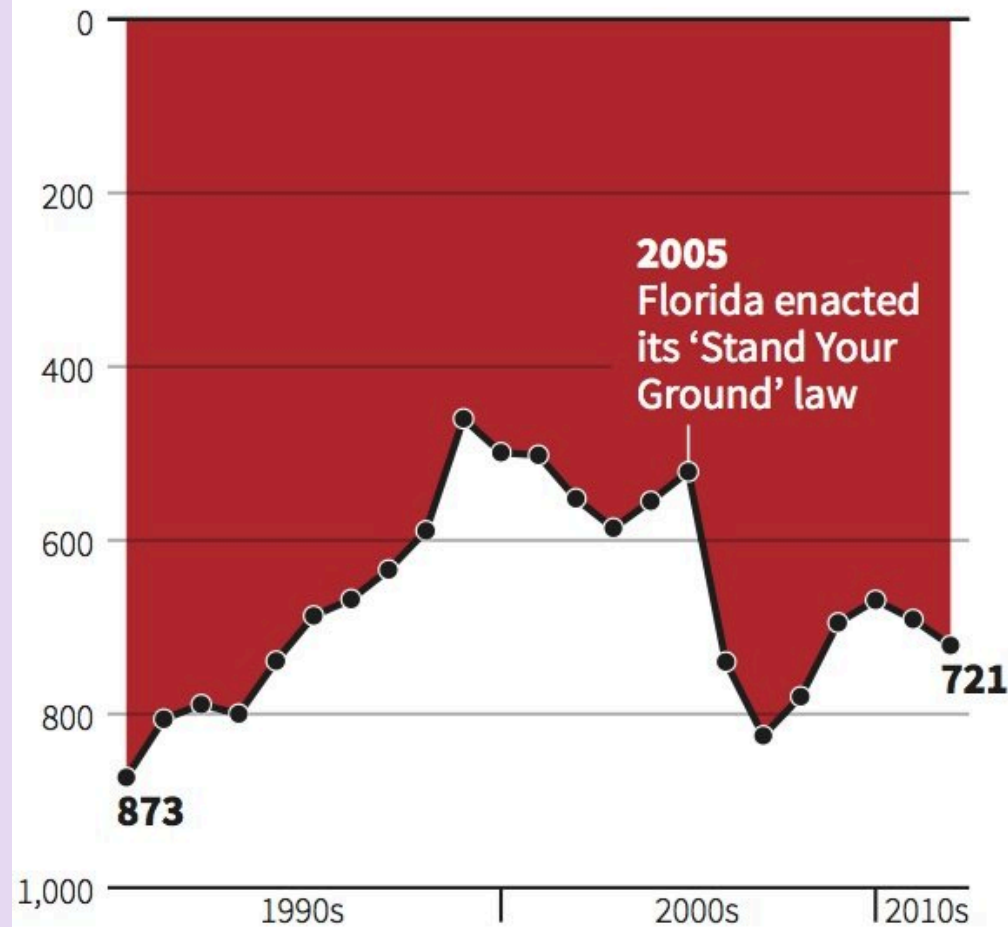


Source: "Yearly Fluctuations in Area of Arctic Covered by Ice" by Derek Watkins (New York Times)



Gun deaths in Florida

Number of murders committed using firearms



Source: Florida Department of Law Enforcement

C. Chan 16/02/2014

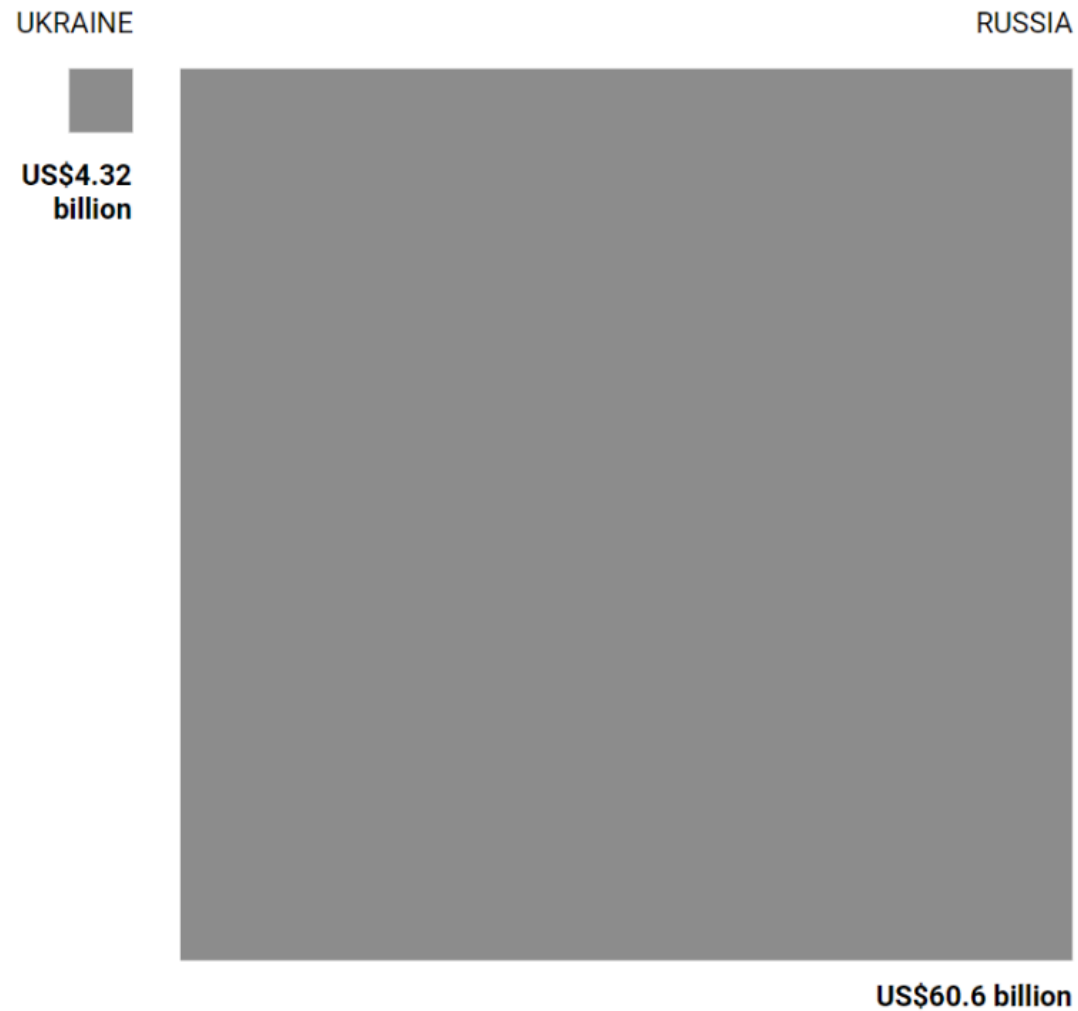
REUTERS

"Gun deaths in Florida" by Reuters Graphics



DEFENCE BUDGETS: RUSSIA VS UKRAINE (2020)

The national balance of forces is overwhelmingly in Russia's favour. Russian military spending in 2020 amounted to US\$60.6 billion in 2020. Ukraine's was less than a 10th of that amount.



"Russia attacks Ukraine" by SCMP Graphic (South China Morning Post)



Work on Your Own Project

Check your own data and consider:

- **Data collection process:** How was the data collected: manually measured, via surveys, automatically recorded, or estimated? Are there potential sources of error?
- **Completeness:** Are there gaps in the data, such as missing values or groups/time periods that were not recorded?
- **Extreme values:** Are there values that appear unrealistically high or low? What could be the cause? How do you generally handle outliers?
- **Level of aggregation:** At what level is your data available? What details might be lost as a result?

⇒ **Discuss problems and potential solutions.**



Exercise

Create a **compelling narrative** for each of two target groups regarding bike rental patterns in London.

- The bike-sharing provider's **business management**
- **Tourists and potential users** of the bike-sharing system

Think about the focus of your narrative and how you can ensure that the audience reaches the desired understanding.



Work on Your Own Project

Define your message and the context:

- **Take-home message:** What is the “big story” behind your visualization? Formulate it in a concise sentence.
- **Communication channel:** How will the communication take place (journal or report, presentation, poster, web, social media)? What specific factors should you consider?
- **Target audience:** For whom are you creating the visualization? Describe your audience regarding their expertise, attitude, and interests.
- **Impact:** What do you want the audience to understand? How should they feel, and what should they do afterwards?
- **Relevance:** List all the variables in your data. Eliminate any that are not essential to the take-home message.

⇒ **Present your main message and selected variables.**



Exercise

Define the story for the survey outcome. Then, explore different ways to present the information.

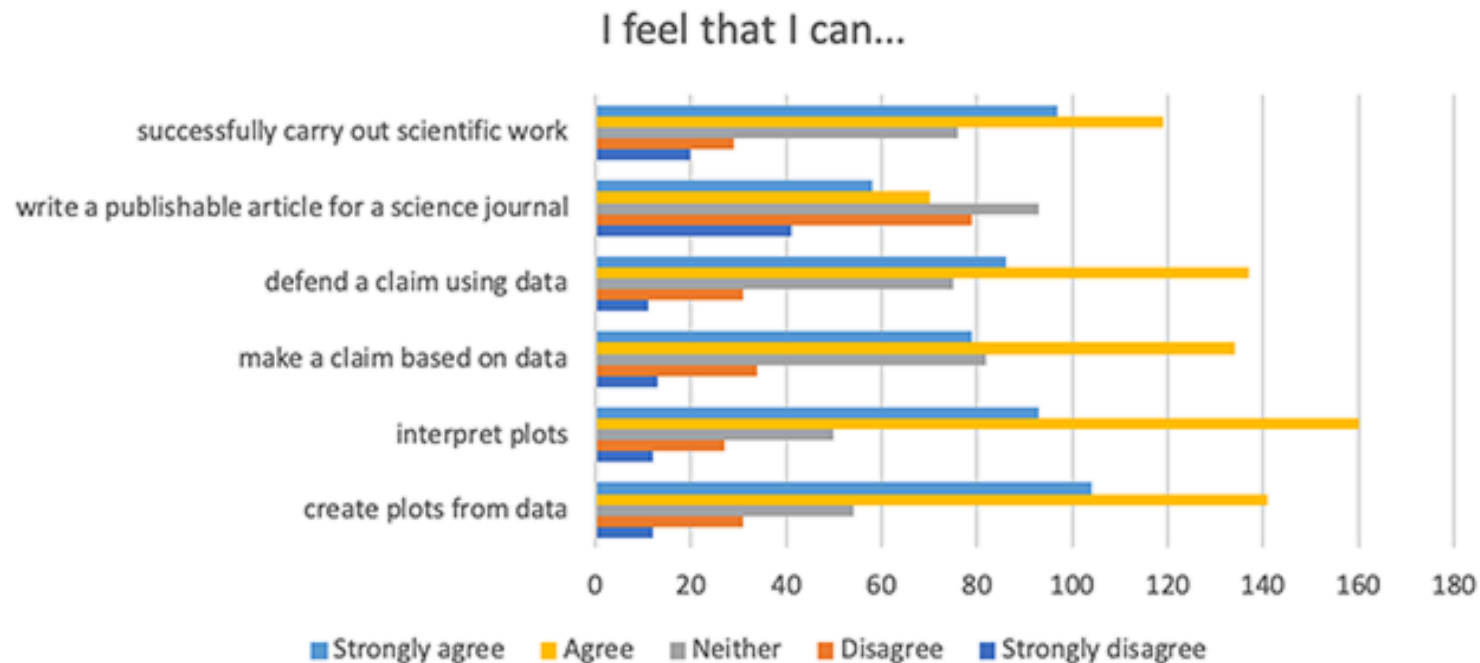


Figure 6

Histogram of responses to questions regarding perceived data analysis and reporting skills (N = 490).



Work on Your Own Project

Decide on the chart type and discuss:

- **Options:** Select several potential chart types that match your data types and the core message.
- **Visualization:** Sketch out possible implementations based on your dataset or create initial visualizations.
- **Variations:** Try out different approaches, e.g., by changing the perspective or level of aggregation.
- **Appropriateness:** Which variant best supports your message and fits the context?

⇒ **Present your preferred chart type and explain your choice.**

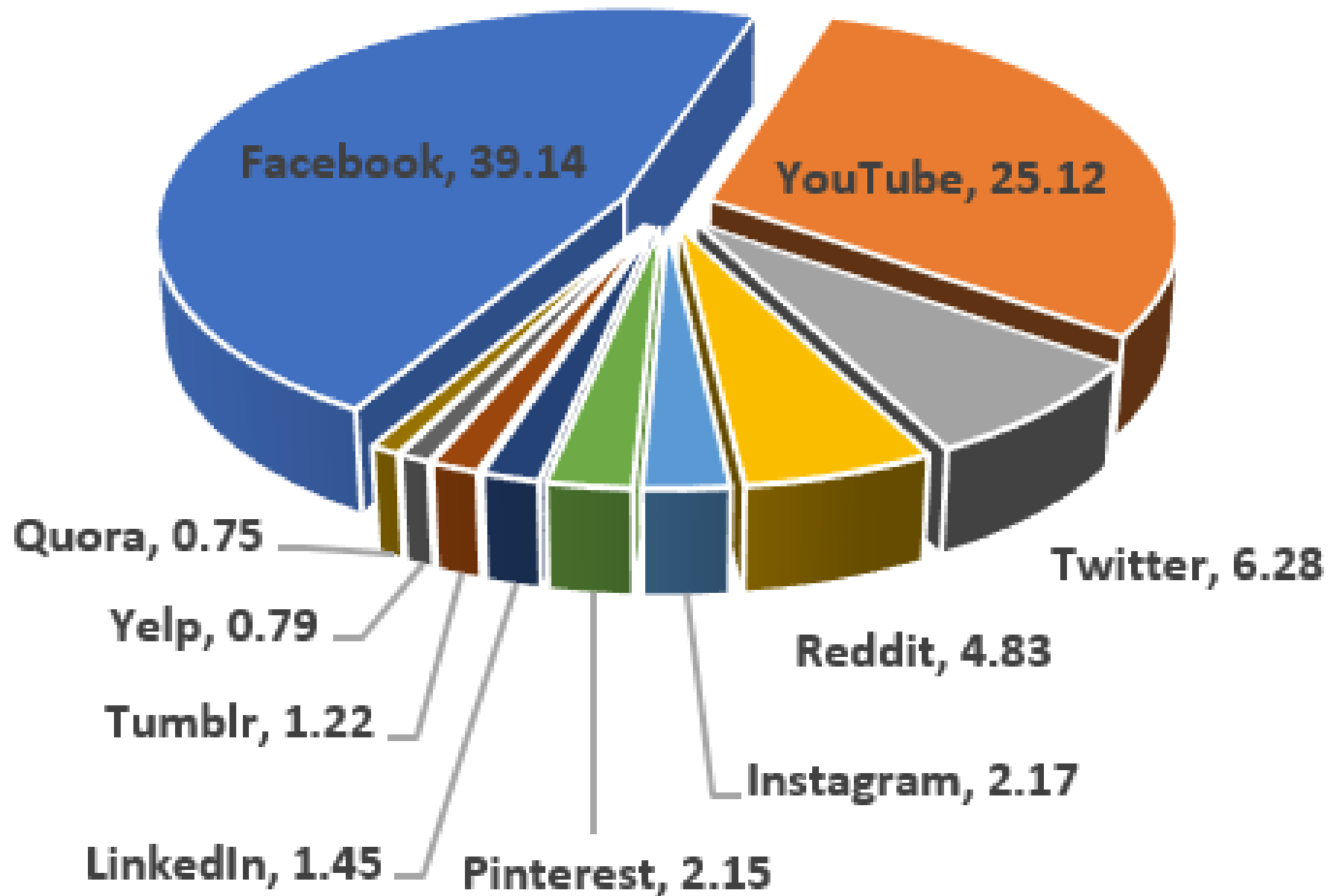


Exercise

Discuss the following chart.

- Are the values shown **plausible**, and is the data **presented accurately**?
- What is the **central message** conveyed by the visualization?
- Is all the data shown **relevant to this message**?
- Does the chosen **chart type** support the message? What works well—and what doesn't?
- What alternative chart type could convey the message clearly and effectively?
Sketch an alternative representation.
- Which chart types would be suitable to **compare the outcomes for two different time periods**?





Market share of visits to social network sites in November 2017,
found in a blog post about [“Why you shouldn’t use pie charts”](#) by The University of Melbourne



Work on Your Own Project

Optimize your visualization and discuss:

- **Focus:** Where does the eye go first? Is the take-home message immediately apparent? Test it with your neighbor.
- **Emphasis:** Consider how you can strategically use pre-attentive attributes (size, position, contrast, color) to support the core message.
- **Palette Type:** Which palette type suits your data: sequential, divergent, or categorical?
- **Accessibility:** Check your visualization for color blindness and contrast issues.

→ **Discuss your highlighting strategy and color choices.**



Exercise

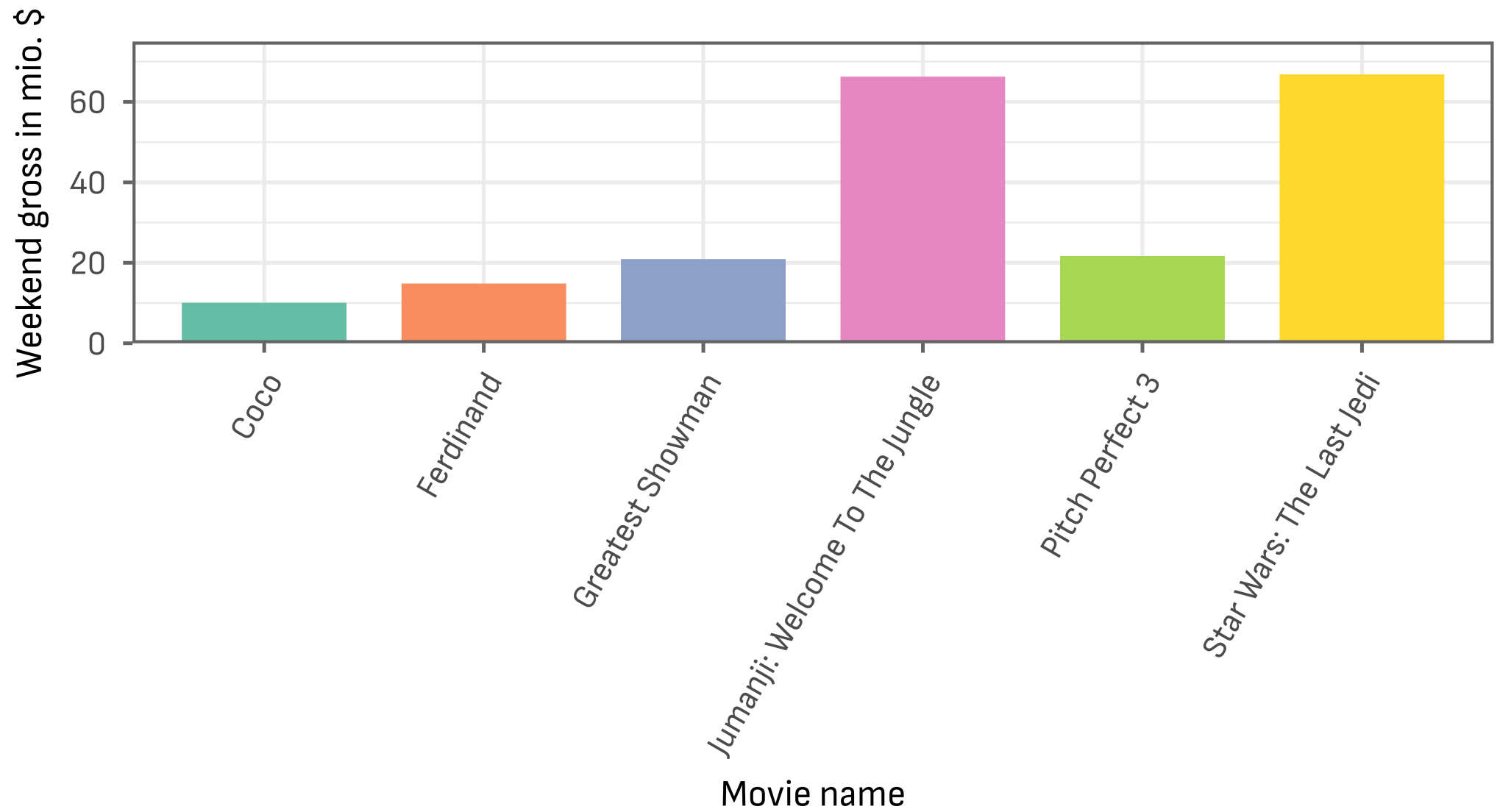
Take a close look at the data visualization comparing the box office receipts during a specific weekend.

- What is the **key message** of the graphic?
- Is all the **information relevant**?
- What **superfluous details** distract from the important data?

Improve the graphic to sharpen the focus:

- Remove **unnecessary elements**.
- Use **pre-attentive features** to highlight the main message.
- Write a **title** that conveys the message.





Work on Your Own Project

Make your graphic clearer and easier to read:

- **Decluttering:** Which elements can you remove or simplify (borders, legends, lines, shapes) without losing information?
- **Clarity:** Is the graphic largely free of decorative elements, and are the most important elements clearly in the foreground?
- **Layout:** Are the text elements easy to read? Could legends and labels be placed closer to the data, and can whitespace be used more effectively?

⇒ List three specific layout changes for the discussion.



Chart Checklist

- Consider whether **errors** during data collection, issues arising from hidden patterns, “guesstimation,” or similar factors might have affected your data.
- Adjust to the **context, tone, medium of communication, and target audience**.
- Use the **“big idea” as the chart title** to guide the development of your visualization.
- Select several **potential chart types** that suit your data types and chosen encodings.
- Experiment with different **variations**, e.g. by changing the perspective or color palette.
- Ensure that the **key message and the data take center stage**.
- Test **which details catch the eye first**, adjust your presentation accordingly.
- Use **pre-attentive attributes and call-outs** to draw attention to what is important.
- Consistently use **intuitive, colorblind-friendly colors** and **sufficient contrast**.
- **Avoid excessive decoration** as well as too many axis labels and grid lines.

